

Lunar Image Preprocessing & Photogrammetry Knowledge Base

Computer Vision Techniques for Illumination-Invariant Feature Matching & RAG Vector Indexing

1. General Image Preprocessing (`image\_preprocessing.txt`)

CONCEPT_NAME: Radiometric Normalization

CATEGORY: Intensity Calibration

METHOD_TYPE: Sensor Calibration & Flat Field Correction

DESCRIPTION: Radiometric normalization converts raw Digital Numbers (DN) collected by line-scan sensors into absolute spectral radiance ($W / m^2 / sr / \mu m$) or surface reflectance (I/F). This removes sensor gain inconsistencies, exposure duration variability, and dark current bias across individual CCD arrays.

ALGORITHMIC_STEPS:

1. Subtract dark current and background bias frames from raw pixel values.
2. Multiply by relative non-uniformity response (flat-field matrix) to correct pixel-to-pixel sensitivity variations.
3. Divide calibrated radiance by solar irradiance (F) normalized by solar distance to compute unitless radiance-over-flux (I/F).

TARGET_METRIC: Radiance Unitless Reflectance (I/F)

COMPUTER_VISION_BENEFIT: Ensures pixel values represent true surface material albedo rather than sensor exposure parameters, stabilizing descriptor extraction across heterogeneous camera settings.

CONCEPT_NAME: Contrast Limited Adaptive Histogram Equalization (CLAHE)

CATEGORY: Contrast Enhancement

METHOD_TYPE: Local Adaptive Histogram Processing

DESCRIPTION: CLAHE improves local contrast in low-light and shaded terrain images by partitioning the image into small contextual regions (tiles) and applying histogram equalization to each tile independently. It limits contrast enhancement by clipping the histogram at a pre-defined threshold to prevent severe noise amplification in homogeneous surface regions.

ALGORITHMIC_STEPS:

1. Divide input image into uniform non-overlapping grid tiles (e.g., 8x8 or 16x16 pixel blocks).
2. Compute intensity histogram for each contextual tile.
3. Clip histogram at a defined limit (e.g., clip_limit = 2.0 to 4.0) and redistribute clipped pixels uniformly across all bins.
4. Apply cumulative distribution function (CDF) transformation to equalize tile histogram.
5. Combine adjacent tiles using bilinear interpolation to remove boundary artifacts between block regions.

TARGET_METRIC: Local Dynamic Range / Feature Contrast

COMPUTER_VISION_BENEFIT: Accurately exposes low-contrast terrain features inside permanently shadowed regions (PSRs) and dimly lit crater floors without saturating adjacent brightly lit crater rims.

CONCEPT_NAME: Gaussian Filtering and Denoising

CATEGORY: Noise Suppression

METHOD_TYPE: Spatial Low-Pass Filtering

DESCRIPTION: Gaussian filtering smooths digital images by convolving the input image matrix with a two-dimensional Gaussian kernel. This operation removes high-frequency shot noise, sensor readout noise, and high-frequency gain fluctuations while suppressing pixel jitter.

ALGORITHMIC_STEPS:

1. Construct a 2D isotropic Gaussian convolution kernel where weight is proportional to distance from center.
2. Convolve image frame with kernel using sliding spatial windows.
3. Optionally substitute Non-Local Means (NL-Means) or Fast Bilateral Filtering to suppress noise while preserving sharp topographical crater edges and ridge lines.

TARGET_METRIC: Signal-to-Noise Ratio (SNR) / Structural Similarity Index (SSIM)

COMPUTER_VISION_BENEFIT: Prevents spurious gradient activations during keypoint detection (e.g., SIFT, HARRIS) caused by high-frequency CCD noise.

CONCEPT_NAME: Edge Enhancement

CATEGORY: Feature Preservation

METHOD_TYPE: Spatial High-Pass Filtering / Unsharp Masking

DESCRIPTION: Edge enhancement amplifies local intensity gradients to sharpen subtle topographic micro-features, impact crater rims, and boulder boundaries.

ALGORITHMIC_STEPS:

1. Generate a low-pass blurred version of the input image using a spatial Gaussian filter.
2. Subtract low-pass image from original image to isolate high-frequency spatial edge mask.
3. Add weighted high-frequency edge mask back to original image matrix using scaling parameter alpha.

TARGET_METRIC: Image Sharpness / Spatial Gradient Magnitude

COMPUTER_VISION_BENEFIT: Increases keypoint repeatability and orientation accuracy for visual descriptors operating under smooth, low-contrast illumination conditions.

2. Radiometric Corrections (`radiometric\_correction.txt`)

CONCEPT_NAME: Flat-Field Correction

CATEGORY: Sensor Calibration

METHOD_TYPE: Multiplicative Optical Sensitivity Correction

DESCRIPTION: Flat-field correction normalizes non-uniform illumination and vignetting artifacts caused by sensor optical lenses, dirt on optics, and gain variation across line-scan detectors.

ALGORITHMIC_STEPS:

1. Acquire uniform reference exposure frame (flat-field frame).
2. Calculate mean pixel value of flat-field matrix.
3. Divide target image by flat-field matrix and multiply by mean pixel value.

TARGET_METRIC: Spatial Gain Uniformity

COMPUTER_VISION_BENEFIT: Eliminates systematic artificial intensity gradients that distort image-wide feature descriptor matching.

CONCEPT_NAME: Dark-Current Subtraction

CATEGORY: Sensor Calibration

METHOD_TYPE: Additive Bias Removal

DESCRIPTION: Dark-current subtraction removes dark noise caused by thermal accumulation of electrons in CCD or CMOS pixels during sensor integration time.

ALGORITHMIC_STEPS:

1. Capture dark frame with camera shutter closed at operational sensor temperature and exposure duration.
2. Subtract dark frame pixel-by-pixel from target image raw frame.

TARGET_METRIC: Absolute Zero Light Baseline

COMPUTER_VISION_BENEFIT: Restores accurate zero-light baseline, ensuring true zero reflectance in total shadow.

CONCEPT_NAME: Spacecraft Jitter Correction

CATEGORY: Geometric Radiometric Artifact Suppression

METHOD_TYPE: Frequency Domain Suppression / Sensor Line Timing Calibration

DESCRIPTION: High-frequency spacecraft vibrations (jitter) create periodic spatial distortion waves across pushbroom sensor scan lines.

ALGORITHMIC_STEPS:

1. Compute Fourier transform across line scan timing sequences.
2. Identify high-frequency jitter periodicities outside expected orbital motion vectors.
3. Apply phase matching and inverse spatial displacement shifts to realign pixel rows.

TARGET_METRIC: Inter-line Spatial Alignment RMS Error

COMPUTER_VISION_BENEFIT: Prevents false feature detection triggered by high-frequency spatial waviness along crater rims and linear topography.

3. Illumination Normalization (`illumination\_normalization.txt`)

CONCEPT_NAME: Photometric Normalization (MMPF / Hapke Model)

CATEGORY: Solar Illumination Normalization

METHOD_TYPE: Semi-Empirical Radiance Normalization

DESCRIPTION: Converts observed image surface reflectance (I/F) acquired under variable solar incidence, emission, and phase angles to a standardized lighting geometry (standard geometry: incidence angle $i = 30$ deg, emission angle $e = 0$ deg, phase angle $g = 30$ deg).

ALGORITHMIC_STEPS:

1. Extract solar incidence (i), emission (e), and phase (g) angles for each pixel from SPICE kernels and DEM surface normal vectors.
2. Compute theoretical surface reflectance using Hapke, Lommel-Seeliger, or Mean Moon Photometric Function (MMPF) photometric models.
3. Divide observed I/F reflectance by model predicted reflectance at actual viewing geometry.
4. Multiply normalized ratio by model predicted reflectance at standard viewing geometry.

TARGET_METRIC: Photometrically Normalized Reflectance (I/F_norm)

COMPUTER_VISION_BENEFIT: Normalizes surface pixel brightness across images acquired at different times of day, enabling matching across vastly different sun positions.

CONCEPT_NAME: Shadow Invariant Feature Transformation

CATEGORY: Illumination Handling

METHOD_TYPE: Color-Space / Intensity Ratio Transformation

DESCRIPTION: Transforms pixel intensities into illumination-invariant representations by computing local intensity gradients or log-reflectance ratios that cancel out global multiplicative light intensity variations.

ALGORITHMIC_STEPS:

1. Apply logarithmic transformation to pixel reflectance values.
2. Compute spatial derivative (gradient) across x and y spatial dimensions.
3. Normalize spatial gradient vectors by local gradient magnitude.

TARGET_METRIC: Illumination-Invariant Spatial Gradient

COMPUTER_VISION_BENEFIT: Maintains high feature matching correlation when matching identical surface locations between morning and afternoon orbits.

4. Shadow Handling (`shadow\_handling.txt`)

CONCEPT_NAME: Shadow Detection and Masking

CATEGORY: Region Segmentation

METHOD_TYPE: Dynamic Thresholding and Geometric Ray-Tracing

DESCRIPTION: Identifies and segments regions of total darkness (cast shadows and self-shadows) where zero direct solar illumination is received, preventing feature matching algorithms from matching uninformative black regions.

ALGORITHMIC_STEPS:

1. Apply absolute radiance thresholding (e.g., $I/F < 0.005$) combined with Otsu adaptive thresholding on localized regions.
2. Validate shadow boundaries by casting solar illumination vectors across high-resolution Digital Elevation Models (DEMs).
3. Generate binary shadow mask (0 for shadowed terrain, 1 for illuminated terrain).

TARGET_METRIC: Binary Shadow Mask Array

COMPUTER_VISION_BENEFIT: Excludes deep shadow boundaries and featureless dark regions from feature detector keypoint selection, reducing false-positive correspondences.

CONCEPT_NAME: Shadow Inpainting and Contrast Restoration

CATEGORY: Image Reconstruction

METHOD_TYPE: Contextual Intensity Transfer / Deep Learning Inpainting

DESCRIPTION: Restores faint secondary scattered light signals inside deep shadowed zones using high-dynamic-range scaling or generative contextual inpainting.

ALGORITHMIC_STEPS:

1. Isolate shadowed regions using shadow mask.
2. Extract low-signal scattered skylight/earthshine radiance within shadowed regions.
3. Apply adaptive non-linear gain curves to boost signal-to-noise ratio in low-signal regions while masking pure dark current noise.

TARGET_METRIC: Dynamic Signal Recovery in Low-Lux Regions

COMPUTER_VISION_BENEFIT: Uncovers hidden surface texture inside partially shaded craters, increasing keypoint density in dark regions.

5. Resolution Normalization (`resolution\_normalization.txt`)

CONCEPT_NAME: Image Pyramids and Scale-Space Normalization

CATEGORY: Multiscale Feature Preprocessing

METHOD_TYPE: Gaussian Scale Space Construction

DESCRIPTION: Constructs a continuous series of progressively downsampled and smoothed image versions to represent surface features across multiple spatial scale levels.

ALGORITHMIC_STEPS:

1. Apply Gaussian smoothing filter with scale parameter sigma to input image.
2. Downsample smoothed image by factor of 2 in both horizontal and vertical directions.
3. Repeat smoothing and downsampling across octave scales to form spatial image pyramid.

TARGET_METRIC: Scale-Space Feature Stability

COMPUTER_VISION_BENEFIT: Enables feature matching algorithms (e.g., SIFT, ORB, SuperPoint) to find correspondences between high-resolution images (0.5 m/px) and lower-resolution contextual images (2.0 m/px).

CONCEPT_NAME: Resampling and Resolution Matching

CATEGORY: Spatial Grid Normalization

METHOD_TYPE: Bicubic / Lanczos Spatial Interpolation

DESCRIPTION: Resamples multi-temporal image grids to a matching target Ground Sample Distance (GSD) prior to feature descriptor computation.

ALGORITHMIC_STEPS:

1. Calculate source image resolution and target common resolution (e.g., reproject 0.72 m/px to exact 1.00 m/px grid).
2. Define transformation grid relative to reference spatial coordinate framework.
3. Apply Lanczos-3 or Bicubic spatial interpolation filter to compute resampled pixel values.

TARGET_METRIC: Uniform Ground Sample Distance (GSD)

COMPUTER_VISION_BENEFIT: Eliminates descriptor orientation and scale mismatch caused by differing orbital altitudes between spacecraft passes.

6. Orthorectification & Registration (`orthorectification.txt`)

CONCEPT_NAME: Photogrammetric Orthorectification

CATEGORY: Geometric Correction

METHOD_TYPE: Sensor Model and DEM Ray-Tracing Projection

DESCRIPTION: Corrects image distortions caused by camera tilt angle, spacecraft position uncertainty, and terrain relief displacement by projecting pixels onto a 3D Digital Elevation Model (DEM) surface.

ALGORITHMIC_STEPS:

1. Load high-precision spacecraft position and pointing orientation vectors from SPICE kernels (CK, SPK, IK).
2. Establish rigorous sensor camera model (rigorous pushbroom interior/exterior orientation).
3. Trace ray path from each image pixel through camera optical center to intersect 3D DEM surface mesh (e.g., LOLA / LROC merged DTM).
4. Project intersected 3D surface coordinates onto target map projection system (e.g., Polar Stereographic or Equirectangular).
5. Interpolate gray values onto regular orthorectified digital orthophoto map (DOM) grid.

TARGET_METRIC: Absolute Geospatial Registration Accuracy (RMS Error < 1.0 Pixel)

COMPUTER_VISION_BENEFIT: Converts raw perspective-distorted images into true map-accurate images where pixel distances correspond directly to ground metric coordinates, allowing direct planar image-to-image feature matching.

CONCEPT_NAME: Co-Registration Preprocessing and Control Network Alignment

CATEGORY: Inter-Image Spatial Alignment

METHOD_TYPE: Bundle Adjustment and Virtual Tie-Point Matching

DESCRIPTION: Co-registers multi-temporal image sets covering identical geographic areas by identifying tie-points across images and applying joint bundle adjustment to remove residual pointing errors.

ALGORITHMIC_STEPS:

1. Extract structurally invariant landmarks across image sets (e.g., crater rim centroids or morphological features).
2. Construct sparse control network linking multi-temporal images.
3. Run global bundle adjustment to update spacecraft trajectory and camera pointing matrices.
4. Apply warping transformation (affine or thin-plate spline) to geometrically align image swaths.

TARGET_METRIC: Relative Inter-Image Displacement Error (Sub-Pixel RMS)

COMPUTER_VISION_BENEFIT: Guarantees that corresponding search windows in multi-temporal images map to the exact same physical ground coordinates before feature matching begins.